Algebra & Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Linear inequalities**

Solve $2x - 5 < 9$ and graph the solution set on a number line.

$2x < 14 \Rightarrow x < 7$.

2 **[Foundational]** • **Target: Solving linear equations**

Solve $3x - 7 = 11$.

$3x = 18$, so $x = 6$.

3 **[Foundational]** • **Target: Solving linear equations**

Solve $5(x - 2) = 3x + 4$.

$5x - 10 = 3x + 4 \Rightarrow 2x = 14 \Rightarrow x = 7$.

4 **[Foundational]** • **Target: Solving linear equations**

Solve $\frac{3x - 2}{4} = x - 1$.

$3x - 2 = 4(x - 1) = 4x - 4 \Rightarrow -x = -2 \Rightarrow x = 2$.

5 **[Foundational]** • **Target: Solving linear equations**

Solve $2(x + 3) - 5 = 3(x - 1)$.

$2x + 1 = 3x - 3 \Rightarrow -x = -4 \Rightarrow x = 4$.

6 **[Foundational]** • **Target: Expanding and simplifying expressions**

Expand and simplify $3(2x - 1) - 2(x - 4)$.

$6x - 3 - 2x + 8 = 4x + 5$.

7 **[Foundational]** • **Target: Expanding and simplifying expressions**

Expand $(x + 5)(x - 2)$.

$x^2 - 2x + 5x - 10 = x^2 + 3x - 10$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Expanding and simplifying expressions

Expand $(2x - 3)^2$.

$(2x - 3)^2 = 4x^2 - 12x + 9$.

9 [Intermediate] • Target: Expanding and simplifying expressions

Simplify $\frac{6x^2 - 9x}{3x}$ (for $x \neq 0$).

$\frac{3x(2x - 3)}{3x} = 2x - 3$.

10 [Intermediate] • Target: Factoring quadratic expressions

Factor $x^2 + 7x + 12$.

$(x + 3)(x + 4)$.

11 [Intermediate] • Target: Factoring quadratic expressions

Factor $x^2 - 2x - 15$.

$(x - 5)(x + 3)$.

12 [Intermediate] • Target: Factoring quadratic expressions

Factor $2x^2 + 5x - 3$.

$(2x - 1)(x + 3)$.

13 [Intermediate] • Target: Factoring quadratic expressions

Factor $x^2 - 49$ using the difference of squares.

$(x - 7)(x + 7)$.

14 [Intermediate] • Target: Solving quadratic equations by factoring

Solve $x^2 - 5x + 6 = 0$.

$(x - 2)(x - 3) = 0$, so $x = 2$ or $x = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 9.01 Statistics & Probability

Solve $x^2 + 2x - 8 = 0$.

$(x + 4)(x - 2) = 0$, so $x = -4$ or $x = 2$.

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16 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 2.01 Functions & Transformations

Solve $2x^2 - 5x - 3 = 0$.

$(2x + 1)(x - 3) = 0$, so $x = -\frac{1}{2}$ or $x = 3$.

17 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 3.01 Sequences & Series

Solve $x^2 - 9x + 20 = 0$.

$(x - 4)(x - 5) = 0$, so $x = 4$ or $x = 5$.

18 [Challenging] • Target: The quadratic formula and the discriminant • Cross-Lesson: 4.01 Trigonometry

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.

$a = 1, b = 3, c = 1$. $x = \frac{-3 \pm \sqrt{9 - 4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$.

19 [Challenging] • Target: The quadratic formula and the discriminant • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.

$\Delta = (-4)^2 - 4(2)(5) = 16 - 40 = -24$. Since $\Delta < 0$, there are no real roots.

20 [Challenging] • Target: The quadratic formula and the discriminant • Cross-Lesson: 6.01 Limits & Derivatives

Find the discriminant of $x^2 - 6x + 9 = 0$ and state how many real roots it has.

$\Delta = (-6)^2 - 4(1)(9) = 36 - 36 = 0$. Since $\Delta = 0$, there is one repeated real root ($x = 3$).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Absolute value equations

The graph shows $y = |x - 3|$. At what x-value does the graph reach its minimum?

a. $x = 1$

b. $x = 3$ ✓

c. $x = -3$

d. $x = 0$

The vertex of the V-shape (the minimum) is at $x = 3$.

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22 [Foundational] • Target: MCQ — Linear inequalities

Solve $3x - 7 < 14$, and identify the graph of its solution set.

a. $x < 7$ ✓

b. $x \geq 7$

c. $x \leq 7$

d. $x > 7$

$3x < 21 \Rightarrow x < 7$.

23 [Foundational] • Target: MCQ — Linear inequalities

Solve $-4x + 6 \geq -10$, and identify the graph of its solution set.

a. $x > 4$

b. $x < 4$

c. $x \leq 4$ ✓

d. $x \geq 4$

$-4x \geq -16$; dividing by -4 flips the sign: $x \leq 4$.

24 [Foundational] • Target: MCQ — Linear-quadratic systems

The graph shows the system $y = x^2 - 4$, $y = 3x$. What are the points of intersection?

a. $(4, 12)$ and $(1, 3)$

b. $(-4, -12)$ and $(1, 3)$

c. $(-1, -3)$ and $(1, 3)$

d. $(4, 12)$ **and** $(-1, -3)$ ✓

The curve and line cross at $(-1, -3)$ and $(4, 12)$.

25 [Foundational] • Target: MCQ — Linear-quadratic systems

The graph shows the system $y = x^2$, $y = 2x + 3$. What are the points of intersection?

a. $(3, 9)$ and $(1, 1)$

b. $(-1, 1)$ and $(1, 1)$

c. $(-3, 9)$ and $(1, 1)$

d. $(3, 9)$ **and** $(-1, 1)$ ✓

The curve and line cross at $(-1, 1)$ and $(3, 9)$.

26 [Foundational] • Target: MCQ — Solving linear equations

Solve $4x - 9 = 15$.

a. $x = 6$ ✓

b. $x = -6$

c. $x = 24$

d. $x = 1.5$

$4x = 24 \Rightarrow x = 6$.

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27 [Foundational] • Target: MCQ — Solving linear equations

Solve $5x + 3 = 2x - 9$.

a. $x = -2$

b. $x = 2$

c. $x = 4$

d. $x = -4$ ✓

$3x = -12 \Rightarrow x = -4$.

28 [Foundational] • Target: MCQ — Factoring quadratic expressions

Factor $x^2 + 9x + 14$.

a. $(x + 2)(x + 7)$ ✓

b. $(x - 2)(x - 7)$

c. $(x + 1)(x + 14)$

d. $(x + 14)(x - 1)$

Need two numbers multiplying to 14, adding to 9: $2, 7$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Factoring quadratic expressions

Factor $2x^2 - x - 6$.

a. $(2x + 3)(x - 2)$ ✓

b. $(2x - 3)(x + 2)$

c. $(x + 3)(2x - 2)$

d. $(2x + 1)(x - 6)$

$(2x + 3)(x - 2) = 2x^2 - 4x + 3x - 6 = 2x^2 - x - 6$.

30 [Intermediate] • Target: MCQ — The quadratic formula and the discriminant

Use the quadratic formula to solve $x^2 + 5x + 2 = 0$, leaving the answer in surd form.

a. $x = \frac{5 \pm \sqrt{17}}{2}$

b. $x = \frac{-5 \pm \sqrt{17}}{4}$

c. $x = \frac{-5 \pm \sqrt{33}}{2}$

d. $x = \frac{-5 \pm \sqrt{17}}{2}$ ✓

Discriminant $= 25 - 8 = 17$; $x = \frac{-5 \pm \sqrt{17}}{2}$.

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31 [Intermediate] • Target: MCQ — Absolute value equations

Solve $|3x + 2| = 13$.

a. $x = \frac{11}{3}$ or $x = 5$

b. $x = 5$ or $x = -5$

c. $x = -\frac{11}{3}$ or $x = 5$

d. $x = \frac{11}{3}$ **or** $x = -5$ ✓

$3x + 2 = 13 \Rightarrow x = \frac{11}{3}$; $3x + 2 = -13 \Rightarrow x = -5$.

32 [Intermediate] • Target: MCQ — Linear inequalities

Solve $2x + 5 > -3$ and $x - 1 < 6$ together (both must hold). Find the combined solution set.

a. $x > -4$

b. $-4 < x \leq 7$

c. $x < 7$

d. $-4 < x < 7$ ✓

$2x + 5 > -3 \Rightarrow x > -4$; $x - 1 < 6 \Rightarrow x < 7$. Together: $-4 < x < 7$.

33 [Intermediate] • Target: MCQ — Rational equations

Solve $\frac{x}{x-3} = 4$.

a. $x = 12$

b. $x = -4$

c. $x = 4$ ✓

d. $x = 3$

$x = 4(x - 3) = 4x - 12 \Rightarrow -3x = -12 \Rightarrow x = 4$.

34 [Intermediate] • Target: MCQ — Rational equations

Solve $\frac{6}{x} = \frac{2}{x-4}$.

a. $x = 12$

b. $x = 4$

c. $x = -6$

d. $x = 6$ ✓

Cross-multiply: $6(x - 4) = 2x \Rightarrow 6x - 24 = 2x \Rightarrow x = 6$.

35 [Intermediate] • Target: MCQ — Linear-quadratic systems

Solve the system $y = x^2 - 4$, $y = 3x$.

a. $(-4, -12)$ and $(1, 3)$

b. $(-1, -3)$ and $(1, 3)$

c. $(4, 12)$ and $(1, 3)$

d. $(4, 12)$ **and** $(-1, -3)$ ✓

$x^2 - 4 = 3x \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0$: $x = 4, y = 12$ or $x = -1, y = -3$.

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36 [Intermediate] • Target: MCQ — Solving quadratic equations by factoring

Solve $4(x+9)=x+36$.

a. 3

b. -2

c. 0 ✓

d. 2

$4x+36=x+36 \Rightarrow 3x=0 \Rightarrow x=0$.

37 [Intermediate] • Target: MCQ — The quadratic formula and the discriminant

Solve $2(x+7)=3x+23$.

a. -11

b. -7

c. -9 ✓

d. 9

$2x+14=3x+23 \Rightarrow -x=9 \Rightarrow x=-9$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Absolute value equations • Cross-Lesson: 4.01 Trigonometry

Solve $5(x-2)=2x-22$.

a. -2

b. -6

c. 4

d. -4 ✓

$5x-10=2x-22 \Rightarrow 3x=-12 \Rightarrow x=-4$.

39 [Challenging] • Target: MCQ — Linear inequalities • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Solve $2(x+7)=x+10$.

a. 4

b. -2

c. -4 ✓

d. -6

$2x+14=x+10 \Rightarrow x=-4$.

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40 [Challenging] • Target: MCQ — Systems of linear equations • Cross-Lesson: 6.01 Limits & Derivatives

Factor $x^2 - 4x - 32$.

a. $(x - 6)(x + 4)$

b. $(x - 8)(x + 4)$ ✓

c. $(x + 8)(x - 4)$

d. $(x - 8)(x - 2)$

$(x - 8)(x + 4)$.

41 [Challenging] • Target: MCQ — Rational equations • Cross-Lesson: 7.01 Integration & Applications

Factor $x^2 + 4x - 32$.

a. $(x + 8)(x - 4)$ ✓

b. $(x + 10)(x - 4)$

c. $(x - 8)(x + 4)$

d. $(x + 8)(x + 6)$

$(x + 8)(x - 4)$.

42 [Challenging] • Target: MCQ — Solving for a variable in a formula • Cross-Lesson: 8.01 Exponentials & Logarithms

Factor $x^2 - 9$.

a. $(x + 3)(x - 3)$

b. $(x - 3)(x + 3)$ ✓

c. $(x - 1)(x + 3)$

d. $(x - 3)(x - 1)$

$(x - 3)(x + 3)$.

43 [Challenging] • Target: MCQ — Linear-quadratic systems • Cross-Lesson: 9.01 Statistics & Probability

Use the quadratic formula to solve $x^2 + 8x + 3 = 0$, leaving the answer in exact form.

a. $\frac{-8 \pm \sqrt{56}}{2}$

b. $\frac{8 \pm \sqrt{52}}{2}$

c. $\frac{-8 \pm \sqrt{52}}{2}$ ✓

d. $\frac{-8 \pm \sqrt{48}}{2}$

$a = 1, b = 8, c = 3$. $x = \frac{-(8) \pm \sqrt{64 - 12}}{2} = \frac{-8 \pm \sqrt{52}}{2}$.

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44 [Challenging] • Target: MCQ — Word problems: setting up and solving equations • Cross-Lesson: 2.01 Functions & Transformations

Find the discriminant of $3x^2 - 4x + 9 = 0$ and state how many real roots it has.

a. 92, no real roots

b. -92, **no real roots** ✓

c. -92, two distinct real roots

d. -92, one repeated real root

$\Delta = (-4)^2 - 4(3)(9) = 16 - 108 = -92$. Since $\Delta < 0$, so there are no real roots.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Sequences & Series

Solve $|x - 9| = 11$.

a. $-2, 20$

b. $22, -2$

c. $20, 0$

d. $20, -2$ ✓

$x - 9 = 11 \Rightarrow x = 20$, or $x - 9 = -11 \Rightarrow x = -2$.

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